Mei Li

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Education

Shanghai Jiao Tong University, Shanghai, China 2023.9 – 2026.3

Master of Engineering in Computer Science and Technology
Supervisor: Prof. Hongtao Lu @ AGI Lab, School of Computer Science GPA: 3.81/4.0
Main courses: Neural Network Theory and Applications (A+), Image Processing and Machine Vision (A+), Intelligent Computing System (A+)

East China Normal University, Shanghai, China 2014.9 – 2018.7

Bachelor of Science in Mathematics and Applied Mathematics GPA: 3.51/4.0
Main cources: Abstract Algebra (A), Complex Analysis (A), Combinatorics (A), Modern Mathematics (A)

(Internship) Nanyang Technological University, Singapore 2025.8 – 2025.12

Supervisor: Prof. Han Yu @ TrustFUL Lab, College of Computing and Data Science

Selected Research Experiences

Precise Optimization for Compound AI Systems (Accepted by ICML 2026) 2025.10 – 2026.5

Keywords: Compound AI System, Optimization, Residual Learning
Textual Gradient-style optimizers (TextGrad) enable gradient-like feedback propagation through Compound AI Systems (CAS) to optimize prompts. However, a fundamental challenge in deep CAS workflows is "Semantic Entanglement," where feedback signals mix local critiques with noisy upstream contexts, leading to attribution ambiguity and optimization collapse. We propose a framework that reformulates textual backpropagation by enforcing Additive Semantic Deltas to preserve an "Identity Highway" for gradient flow. We introduce a Semantic Projector to decompose entangled gradients into causally independent subspaces, enabling precise signal routing and density-aware scheduling.

Behavior Consolidation: Context-to-Weight Procedural Memory for Agents 2026.5 – Now

Keywords: Agents, Procedural Memory, LoRA, Context Distillation
Behavior Consolidation (BC) aims to convert useful behaviors elicited by long session contexts into compact, detachable LoRA adapters. In agent workflows, repeated reading of logs, prior failures, user preferences, and workflow conventions creates substantial token and latency overhead. We formulate BC as learning the behavior delta between a base model with session context and the same model without context, enabling the adapted model to reproduce context-conditioned behaviors with fewer prompt tokens. We further design synthetic session benchmarks and rule-based evaluation protocols to measure behavior consistency, token savings, and reversibility after unloading the adapter.

Continual Learning with Adaptive Model Merging (Accepted by ICML 2025) 2024.6 – 2025.1

Keywords: Continual Learning, Catastrophic Forgetting, Model Merging
Continual Learning (CL) aims to progressively acquire knowledge from sequential tasks while preventing catastrophic forgetting. A fundamental challenge in CL involves achieving an optimal balance between stability (preserving previously learned knowledge) and plasticity (adapting to new tasks). We investigate how model merging can improve this stability-plasticity trade-off and offer theoretical analysis that highlights its advantages and verify our method on several popular benchmarks.

Semi-supervised Orientation Estimation of Objects (Accepted by CVIU) 2024.5 – 2024.9

Keywords: Orientation Estimation, Semi-supervised Learning

Semi-Supervised Orientation Estimation aims to predict 3D orientations of objects from 2D images using only a limited set of labeled data, leveraging abundant unlabeled samples to improve learning efficiency. We investigate the effectiveness of curriculum-guided pseudo-labeling, enhancing label reliability and improving semi-supervised rotation regression performance.

Publications

1. **BECAME: BayEsian Continual Learning with Adaptive Model MErging**
   Mei Li, Yuxiang Lu, Qinyan Dai, Suizhi Huang, Yue Ding, Hongtao Lu
   International Conference on Machine Learning. **(ICML 2025)**
2. **TextResNet: Decoupling and Routing Optimization Signals in Compound AI Systems via Deep Residual Tuning**
   Suizhi Huang, **Mei Li**, Han Yu, Xiaoxiao Li
   International Conference on Machine Learning. (ICML 2026)
3. **HACMatch: Semi-Supervised Rotation Regression with Hardness-Aware Curriculum Pseudo Labeling**
   Mei Li, Huayi Zhou, Suizhi Huang, Yuxiang Lu, Yue Ding, Hongtao Lu
   Computer Vision and Image Understanding. (CVIU)
4. **A Comprehensive Survey on Benchmarks and Solutions in Software Engineering of LLM-Empowered Agentic System**
   Jiale Guo*, Suizhi Huang*, **Mei Li***, Dong Huang, Xingsheng Chen, Regina Zhang, Zhijiang Guo, Han Yu, Siu-Ming Yiu, Christian Jensen, Pietro Lio, Kwok-Yan Lam (co-first author)
   Under Review. (ACM Computing Surveys)
5. **From Anchors to Answers: A Novel Node Tokenizer for Integrating Graph Structure into Large Language Models**
   Yanbiao Ji, Chang Liu, Xin Chen, Dan Luo, **Mei Li**, Yue Ding, Wenqing Lin, Hongtao Lu
   Conference on Information and Knowledge Management. **(CIKM 2025)**
6. **MORE: Multi-Organ Medical Image REconstruction Dataset**
   Shaokai Wu, Yapan Guo, Yanbiao Ji, Jing Tong, Yue Ding, Yuxiang Lu, **Mei Li**, Suizhi Huang, Hongtao Lu
   ACM International Conference on Multimedia. (**ACMMM 2025** Datasets)

Honors and Awards

National Scholarship of Graduate (top 2% national-wide)	2025.12
Merit Student of Shanghai Jiao Tong University	2023-2024
First-Class Master's Academic Scholarship	2023
Excellent Student Scholarship	2015, 2016, 2017

Services

2025 - 2026 CVPR, IJCAI, ICCV, NuerIPS, TCSVT Reviewer
2025.9 - 2026.3 Class President
2025.09 - 2026.01, 2024.09 - 2025.01 TA, Computer Vision and Image Processing, SCS, SJTU

Skills

Programming Languages: Python, C/C++
Development: PyTorch, OpenCV, Git, CUDA
English Proficiency: IELTS 7.0 / 9.0; GRE 327 (Verbal 157, Quantitative 170) + 4.0 (Writing)